

|                                            |          |
|--------------------------------------------|----------|
| <b>I. Generalities of Smooth Manifolds</b> | <b>1</b> |
| <b>1 Smooth Manifold . . . . .</b>         | <b>1</b> |
| Alphabetical Index                         |          |
| References                                 |          |

## 1 Smooth Manifold

**Definition 1.1.** A topological space  $M$  is a *topological  $n$ -manifold* if the following properties hold.

1. For every  $x \in M$ , there exists a neighborhood  $U_x$  of  $x$  homeomorphic to an open subset of  $\mathbb{R}^n$ .
2. The space  $M$  is second-countable.
3. The space  $M$  is Hausdorff.

**Remark.** The empty set  $\emptyset$  is vacuously a topological  $n$ -manifold. We shall ignore this case.

**Proposition 1.1.** Let  $M$  be a topological space. Then the following are equivalent.

1. For every  $x \in M$ , there exists a neighborhood  $U_x$  of  $x$  homeomorphic to an open subset of  $\mathbb{R}^n$ .
2. For every  $x \in M$ , there exists a neighborhood  $U_x$  of  $x$  homeomorphic to an open ball of  $\mathbb{R}^n$ .
3. For every  $x \in M$ , there exists a neighborhood  $U_x$  of  $x$  such that  $U_x \approx \mathbb{R}^n$ .

*Proof.*  $(2 \Leftrightarrow 3)$  Any open ball in  $\mathbb{R}^n$  is homeomorphic to  $\mathbb{R}^n$ .  $(1 \Rightarrow 3)$  Fix  $x \in M$ . Let  $\varphi : U_x \rightarrow V$  be the homeomorphism. Then there exists an open ball centered at  $\varphi(x)$  of sufficiently small radius such that  $B(\varphi(x)) \subset V$ . Let  $U' = \varphi^{-1}(B(\varphi(x)))$ . Since  $\varphi$  is continuous,  $U'$  is the desired neighborhood of  $x$ .  $(3 \Rightarrow 1)$  Trivial.  $\square$

**Proposition 1.2.** If  $m \neq n$ , then any topological  $m$ -manifold is not homeomorphic to any topological  $n$ -manifold.

A self-contained proof will be shown later.

**Definition 1.2.** Let  $M$  be a topological  $n$ -manifold. A *chart* on  $M$  is a pair  $(U, \varphi)$  where  $U \subset M$  is open and  $\varphi : U \rightarrow V$  is a homeomorphism onto an open subset  $V \subset \mathbb{R}^n$ .

For every  $p \in M$ , there exists a chart  $(U, \varphi)$  such that  $p \in U$ . If  $\varphi(p) = 0$ , then we say  $(U, \varphi)$  is *centered at  $p$* .

**Definition 1.3.** Let  $(U, \varphi)$  be a chart on a topological manifold  $M$ . The set  $U$  is called a *coordinate domain*. If  $\varphi(U)$  is an open ball, then  $U$  is called a *coordinate ball*. If  $\varphi(U)$  is an open cube, then  $U$  is called a *coordinate cube*. The map  $\varphi$  is called the *coordinate map*. For every  $p \in U$ , rewrite  $\varphi(p) = (x_1(p), \dots, x_n(p))$ . The coordinates  $x_i(p)$  are called *local coordinates* on  $U$ .

**Example.** Let  $U \subset \mathbb{R}^n$  be open and let  $f : U \rightarrow \mathbb{R}^k$  be continuous. The *graph* of  $f$  is  $\Gamma(f) = \{(x, y) \mid x \in U, y = f(x)\}$  endowed with the subspace topology, so it is Hausdorff and second countable. Let  $\pi : \mathbb{R}^n \times \mathbb{R}^k \rightarrow \mathbb{R}^n$  be the natural projection, then  $\pi|_{\Gamma(f)}$  is a homeomorphism onto its image. Hence  $\Gamma(f)$  is a topological  $n$ -manifold.  $\bowtie$

**Example.** Recall that  $S^n = \{x \in \mathbb{R}^{n+1} \mid |x| = 1\}$ . Equip  $S^n$  with the subspace topology, then  $S^n$  is Hausdorff and second-countable. Let  $U_i^+ = \{(x_1, \dots, x_{n+1}) \mid x_i > 0\} \subset \mathbb{R}^{n+1}$  and similarly define  $U_i^-$ , so  $\{U_i^\pm\}$  covers  $S^n$ . Define  $f : B^n \rightarrow \mathbb{R}$  by  $f : u \mapsto \sqrt{1 - |u|^2}$ , so  $f$  is continuous. If  $(x_1, \dots, x_{n+1}) \in S^n$ , then  $(x_1, \dots, \widehat{x_i}, \dots, x_{n+1}) \in B^n$ . We have  $\Gamma(f) = U_i^+ \cap S^n$  and  $\Gamma(-f) = U_i^- \cap S^n$  for some  $U_i$ . Since  $\Gamma(f)$  and  $\Gamma(-f)$  are topological  $n$ -manifolds,  $S^n$  is a topological  $n$ -manifold.  $\bowtie$

**Example.** The  $n$ -dimensional *real projective space*  $\mathbb{RP}^n$  is the set of all 1-dimensional linear subspaces of  $\mathbb{R}^{n+1}$ . Equivalently, we may define  $\mathbb{RP}^n$  via the quotient map  $\pi : \mathbb{R}^{n+1} \setminus \{0\} \rightarrow \mathbb{RP}^n$  defined by  $\pi : x \mapsto \langle x \rangle$ . Define  $\widetilde{U}_i = \{(x_1, \dots, x_{n+1}) \mid x_i \neq 0\} \subset \mathbb{R}^{n+1} \setminus \{0\}$  and define  $U_i = \pi(\widetilde{U}_i)$ . Fix  $i$ . Trivially  $\widetilde{U}_i$  is open and saturated with respect to  $\pi$ . The restriction of a quotient map to an open saturated set is continuous, so  $\pi|_{\widetilde{U}_i}$  is a continuous quotient map. Now define  $\varphi_i : U_i \rightarrow \mathbb{R}^n$  by  $\varphi_i : \langle (x_1, \dots, x_{n+1}) \rangle \mapsto (x_1/x_i, \dots, \widehat{x_i/x_i}, \dots, x_{n+1}/x_i)$ . This map is well-defined. The map  $\varphi_i \circ \pi$  on each coordinate is continuous, hence  $\varphi_i \circ \pi$  is continuous on  $\widetilde{U}_i$ . By the universal property for quotient spaces,  $\varphi_i$  is continuous.

$$\begin{array}{ccc}
 \mathbb{R}^n & \xleftarrow{\varphi_i} & U_i \\
 & \nwarrow \varphi_i \circ \pi & \uparrow \pi \\
 & & \widetilde{U}_i
 \end{array}$$

Consider  $\psi_i : \mathbb{R}^n \rightarrow U_i$  defined by  $\psi_i : (y_1, \dots, y_n) \mapsto \langle y_1, \dots, y_{i-1}, 1, y_i, \dots, y_n \rangle$ , and  $\psi_i$  is the continuous inverse of  $\varphi_i$ . Since  $\{U_i\}$  is an open cover of  $\mathbb{RP}^n$ ,  $\mathbb{RP}^n$  is locally Euclidean of dimension  $n$ . Pick  $\langle x \rangle, \langle y \rangle \in \mathbb{RP}^n$ . Suppose  $\langle x \rangle \in U_i$  and  $\langle y \rangle \in U_j$ . Since  $U_i \approx \mathbb{R}^n$ ,  $U_i$  is Hausdorff, so the case  $i = j$  is trivial. Let  $i \neq j$ . Let  $r = \min\{d(x/|x|, y/|y|), d(x/|x|, -y/|y|)\}/3$ . Then  $\langle x \rangle \in \pi(B_r(x/|x|))$ ,  $\langle y \rangle \in \pi(B_r(y/|y|))$ , and  $\pi(B_r(x/|x|)) \cap \pi(B_r(y/|y|)) = \emptyset$ . Hence  $\mathbb{RP}^n$  is Hausdorff. Since each  $U_i$  is second-countable and  $\{U_i\}$  is a finite open cover of  $\mathbb{RP}^n$ ,  $\mathbb{RP}^n$  is second-countable. Hence  $\mathbb{RP}^n$  is a topological  $n$ -manifold.  $\oslash$

**Proposition 1.3.** The space  $\mathbb{RP}^n$  is compact.

*Proof.* For any  $\langle x \rangle \in \mathbb{RP}^n$ ,  $|x/|x|| = 1 \in S^n$ , so  $\pi|_{S^n}$  is surjective. Since  $S^n$  is compact,  $\mathbb{RP}^n$  is compact.  $\square$

**Example.** Let  $M_1, \dots, M_k$  be topological manifolds with  $\dim M_i = n_i$ . The product space  $M = M_1 \times \dots \times M_k$  is Hausdorff and second-countable. For every  $(p_1, \dots, p_k) \in M$ , there exist charts  $(U_i, \varphi_i)$  in each  $M_i$ . Define  $\varphi : U_1 \times \dots \times U_k \rightarrow \varphi_1(U_1) \times \dots \times \varphi_k(U_k)$  by  $\varphi : (x_1, \dots, x_k) \mapsto (\varphi_1(x_1), \dots, \varphi_k(x_k))$  and the image is open in  $\mathbb{R}^{\sum_1^k n_i}$ . Since each  $\varphi_i$  is a homeomorphism,  $\varphi$  is a homeomorphism. Hence  $M$  is a topological  $(\sum_1^k n_i)$ -manifold.  $\oslash$

**Example.** The  $n$ -torus is defined to be the product space  $T^n = S^1 \times \dots \times S^1$  of  $n$  copies of  $S^1$ . Hence  $T^n$  is a topological  $n$ -manifold.  $\oslash$

**Proposition 1.4.** Every topological manifold  $M$  has a countable basis of precompact coordinate balls.

*Proof.* Let  $\dim M = n$ . Since  $M$  is second countable,  $M$  admits a countable cover  $\{U_i\}$ . Fix a chart  $(U_i, \varphi_i)$ . Consider the collection of balls  $B_r(x)$ , where  $r \in \mathbb{Q}$  and  $x \in \mathbb{Q}^n$ , such that  $\overline{B_r(x)} \subset \varphi_i(U_i)$ , then  $\{B_r(x)\}$  is a precompact basis for  $\varphi_i(U_i)$ . The collection  $\{\varphi_i^{-1}(B_r(x))\}$  is a countable basis of  $U_i$ . Since  $\overline{B_r(x)}$  is compact in  $\mathbb{R}^n$ ,  $\varphi_i^{-1}(\overline{B_r(x)})$  is compact in  $U_i$ . Since  $M$  is Hausdorff, the closure of  $\varphi_i^{-1}(B_r(x))$  in  $U_i$  is its closure in  $M$ . Hence the union of  $\{\varphi_i^{-1}(B_r(x))\}$  is a precompact basis for  $M$ .  $\square$

**Proposition 1.5.** Let  $M$  be a topological manifold. Then the following properties hold.

1. The manifold  $M$  is locally path-connected.
2. The manifold  $M$  is connected if and only if it is path-connected.
3. The components of  $M$  are the same as its path components.
4. The manifold  $M$  has countably many components and each of which is an open subset of  $M$  and a connected topological manifold.

*Proof.* (i) Open balls are path-connected. Precompact coordinate balls are path-connected, hence  $M$  is locally path-connected. (ii) Path-connected spaces are connected. By (i),  $M$  is locally path-connected. Since  $M$  is connected,  $M$  is path-connected. (iii) Since  $M$  is locally path-connected, the path-components are open. Path-components partition  $M$ , so they are clopen. Every path-component is a subset of a connected component. Since every component is connected, it cannot be partitioned into multiple clopen subsets, so each component contains exactly one path-component. (iv) Every component of  $M$  is open. All components cover  $M$ , so there exists a countable subcover. Since components are disjoint, there are countably many components. Each component is a subspace of  $M$ , so it is second-countable and Hausdorff, hence a connected topological manifold.  $\square$

**Proposition 1.6.** Every topological manifold is locally compact.

*Proof.* A topological manifold is Hausdorff and admits a basis of precompact sets, hence it is locally compact.  $\square$

**Definition 1.4.** Let  $M$  be a topological space. Let  $X \subset \mathcal{P}(M)$ . We say  $X$  is *locally finite* if each  $p \in M$  has a neighborhood that intersects at most finitely many sets in  $X$ .

**Proposition 1.7.** Let  $M$  be a topological space and let  $X$  be locally finite. Then the following properties hold.

1. The set  $\{\overline{X_i} \mid X_i \in X\}$  is locally finite.
2. The set  $\overline{\bigcup X_i} = \bigcup \overline{X_i}$ .

*Proof.* (i) Fix  $p \in M$ . Since  $X$  is locally finite, there exists neighborhood  $U$  of  $p$  such that it intersects  $X_1, \dots, X_n \in X$ . Suppose there exists  $y \in U \cap \overline{X_m}$ , then any neighborhood of  $y$  intersects  $X_m$ , so  $X_m$  is one of  $X_1, \dots, X_n$ , hence  $U$  intersects at most  $n$  closures. (ii) Trivially  $\bigcup \overline{X_i} \subset \overline{\bigcup X_i}$ . Let  $p \in \overline{\bigcup X_i}$  and suppose  $p \notin \bigcup \overline{X_i}$ . Since  $X$  is locally finite, there exists  $U_p$  that only intersects  $X_1, \dots, X_n$ . Since  $p \notin \bigcup \overline{X_i}$ ,  $W = U_p \cap \bigcap_1^n (M \setminus \overline{X_k})$  is a neighborhood of  $p$ . Since  $W \subset U_p$  and  $W \subset M \setminus \overline{X_k}$ ,  $W \cap X_k = \emptyset$ , so  $W \cap \bigcup X_i = \emptyset$ , a contradiction.  $\square$

**Definition 1.5.** Let  $M$  be a topological space and let  $U$  be a cover of  $M$ . A cover  $V$  of  $M$  is a *refinement* of  $U$  if for each  $V_i \in V$ , there exists a  $U_j \in U$  such that  $V_i \subset U_j$ . We say  $M$  is *paracompact* if every open cover of  $M$  admits an open locally finite refinement.

**Proposition 1.8.** Let  $M$  be a topological manifold. For every basis  $B$  of  $M$  and open cover  $X$ , there exists a countable locally finite refinement of  $X$  that consists of basis elements in  $B$ .

*Proof.* Let  $\{K_j\}$  be an exhaustion of  $M$  by compact sets. Define  $V_j = K_{j+1} \setminus \text{Int}(K_j)$  and  $W_j = \text{Int}(K_{j+2}) \setminus K_{j-1}$ , then  $V_j \subset W_j$  is compact. For every  $x \in V_j$ , there exists  $U_x \in X$  with  $x \in U_x$ . Since  $B$  is a basis, there exists  $B_x \in B$  with  $x \in B_x$  such that  $B_x \subset U_x \cap W_j$ . The collection  $\{B_x\}$  is an open cover of  $V_j$ , since  $V_j$  is compact, there exists a finite subcover  $\{B_{x_j}\}$ . The union  $\bigcup \{B_{x_j}\}$  is therefore a countable refinement of  $X$ . Notice that  $W_i$  intersects  $W_j$  if and only if  $|i - j| \leq 2$ , so  $\bigcup \{B_{x_j}\}$  is locally finite.  $\square$

**Proposition 1.9.** The fundamental group of a topological manifold is countable.

*Proof.* Let  $M$  be a topological manifold. Pick a countable basis  $B$  of coordinate balls of  $M$ . For every  $B_1, B_2 \in B$ ,  $B_1 \cap B_2$  has at most countably many path-components. For every path-component, fix a point  $x_i$ , and let  $X$  be the collection of those  $x_i$ . Since  $B$  is countable,  $X$  is countable. For every  $x, y \in X$  and  $U \in B$ , fix a path  $h_{x,y}^U$  joining  $x$  and  $y$  that entirely lies in  $U$ . Let  $\Omega$  be the set of all finite concatenation of such paths, so  $\Omega$  is also countable. Given any loop  $f : [0, 1] \rightarrow M$  based at  $p \in M$ , the set  $\{f^{-1}(U) \mid U \in B\}$  is an open cover of  $[0, 1]$ . Since  $[0, 1]$  is compact, by the Lebesgue's number lemma, there exists a partition  $0 = a_0 < a_1 < \dots < a_k = 1$  such that each  $[a_{i-1}, a_i]$  is mapped by  $f$  entirely into some  $U_i \in B$ . For each  $i \in \{1, \dots, k-1\}$ ,  $f(a_i)$  lies in  $U_i \cap U_{i+1}$ , so  $f(a_i)$  belongs to a specific path-component of  $U_i \cap U_{i+1}$ . Let  $x_i \in X$  be the chosen reference point for that path-component, and let  $g_i$  be a path from  $f(a_i)$  to  $x_i$  lying entirely within that path-component. Define  $x_0 = x_k = p$  and define  $g_0 = g_k = c_p$ , the constant path. Let  $f_i = f|_{[a_{i-1}, a_i]}$ , then  $f \simeq (f_1 * g_1) * (g_1^{-1} * f_2 * g_2) * \dots * (g_{k-1}^{-1} * f_k)$ . Notice that  $g_{i-1}^{-1} * f_i * g_i$  lies in  $U_i$ , since  $U_i$  is simply connected, any paths with the same endpoints are homotopic. Hence  $f$  is homotopic to a path in  $\Omega$  and  $[f] \in \Omega$ . Since  $\Omega$  is countable,  $\pi_1(M, p)$  is countable.  $\square$

**Definition 1.6.** Let  $U \subset \mathbb{R}^n$  and  $V \subset \mathbb{R}^m$  be open. A function  $F : U \rightarrow V$  is a *smooth function* if each component function of  $F$  has continuous partial derivatives of all orders. A bijective smooth map is a *diffeomorphism* if its inverse map is smooth.

We sometimes use  $C^\infty$  *functions* to denote smooth functions.

**Definition 1.7.** Let  $M$  be a topological  $n$ -manifold. Let  $(U, \varphi)$  and  $(V, \psi)$  be charts. The *transition map* from  $\varphi$  to  $\psi$  is the map  $\psi \circ \varphi^{-1} : \varphi(U \cap V) \rightarrow \psi(U \cap V)$  defined by  $\psi \circ \varphi^{-1} : \varphi(x) \mapsto \psi(x)$ . Two charts are *smoothly compatible* if the transition map is a diffeomorphism.

$$\begin{array}{ccc}
 & U \cap V & \\
 \varphi \swarrow & & \searrow \psi \\
 \varphi(U \cap V) & \xleftrightarrow[\psi \circ \varphi^{-1}]{\varphi \circ \psi^{-1}} & \psi(U \cap V)
 \end{array}$$

If  $U \cap V = \emptyset$ , the transition map  $\psi \circ \varphi^{-1}$  is vacuously a diffeomorphism.

**Definition 1.8.** Let  $M$  be a topological manifold. An *atlas*  $\mathcal{A}$  for  $M$  is a collection  $\{(U_i, \varphi_i)\}$  of charts such that  $\bigcup U_i = M$ . An atlas is a *smooth atlas* if any two charts in the collection are smoothly compatible. A smooth atlas  $\mathcal{A}$  for  $M$  is *maximal* if it is not properly contained in any smooth atlas.

**Definition 1.9.** Let  $M$  be a topological manifold. A maximal smooth atlas  $\mathcal{A}$  for  $M$  is called a *smooth structure* on  $M$ . A pair  $(M, \mathcal{A})$  is called a *smooth manifold*.

**Proposition 1.10.** Let  $M$  be a topological manifold. Then the following properties hold.

1. Every smooth atlas  $\mathcal{A}$  for  $M$  is contained in a unique maximal smooth atlas, called the smooth structure determined by  $\mathcal{A}$ .
2. Two smooth atlases for  $M$  determine the same smooth structure if and only if their union is a smooth atlas.

*Proof.* (i) Let  $\mathcal{A}$  be a smooth atlas for  $M$ . Consider the collection  $\bar{\mathcal{A}}$  of all charts that are compatible with every chart in  $\mathcal{A}$ . Clearly,  $\mathcal{A}$  is contained in  $\bar{\mathcal{A}}$ , so  $\bar{\mathcal{A}}$  is an atlas for  $M$ . Let  $(U, \varphi)$  and  $(W, \theta)$  be two charts in  $\bar{\mathcal{A}}$ . For every  $x \in U \cap W$ , there exists a chart  $(V, \psi) \in \mathcal{A}$  that contains  $x$ . On the intersection  $U \cap V \cap W$ , the transition map  $\varphi \circ \psi^{-1} \circ (\psi \circ \theta^{-1}) = \varphi \circ \theta^{-1}$  is a composition of diffeomorphisms, hence a diffeomorphism. Since the map is smooth on  $U \cap V \cap W$ , it is smooth at  $x$ . Because  $x$  was arbitrary, the map is smooth everywhere on  $U \cap W$ . Hence,  $\bar{\mathcal{A}}$  is a smooth atlas. If  $\bar{\mathcal{A}} \subset \mathcal{B}$ , where  $\mathcal{B}$  is a smooth atlas, then any chart in  $\mathcal{B}$  is compatible with all charts in  $\mathcal{A}$ . Hence  $\mathcal{B} \subset \bar{\mathcal{A}}$  and  $\bar{\mathcal{A}}$  is maximal. The uniqueness of  $\bar{\mathcal{A}}$  is trivial. (ii) Let  $\mathcal{B}$  be a smooth atlas for  $M$ . ( $\Rightarrow$ ) If  $\bar{\mathcal{B}} = \bar{\mathcal{A}}$ , then charts in  $\mathcal{B}$  are compatible with charts in  $\mathcal{A}$ , so  $\mathcal{A} \cup \mathcal{B}$  is smooth. ( $\Leftarrow$ ) If  $\mathcal{A} \cup \mathcal{B}$  is smooth, then every chart in  $\mathcal{A}$  is compatible with all charts in  $\mathcal{B}$ , so  $\mathcal{A} \subset \bar{\mathcal{B}}$  and  $\bar{\mathcal{A}} \subset \bar{\mathcal{B}}$ . Similarly,  $\bar{\mathcal{B}} \subset \bar{\mathcal{A}}$ . Hence  $\bar{\mathcal{A}} = \bar{\mathcal{B}}$ .  $\square$

If all transition maps are  $k$ -differentiable ( $C^k$ ) diffeomorphisms instead of  $C^\infty$  diffeomorphisms, we obtain a  $C^k$  structure. A manifold with a  $C^k$  structure is called a  $C^k$  manifold.

**Definition 1.10.** Let  $(M, \mathcal{A})$  be a smooth manifold. Every chart  $(U, \varphi) \in \mathcal{A}$  is called a *smooth chart*. The set  $U$  is called a *smooth coordinate domain* and the map  $\varphi$  is called a *smooth coordinate map*. If  $\varphi(U)$  is a ball, then  $U$  is called a *smooth coordinate ball*.

**Definition 1.11.** Let  $M$  be a smooth  $n$ -manifold. A set  $B \subset M$  is a *regular coordinate ball* if there exists a smooth chart  $(B', \varphi)$  with  $\bar{B} \subset B'$  such that for some positive reals  $r < r'$ ,  $\varphi(B) = B_r(0)$ ,  $\varphi(\bar{B}) = \overline{B_r(0)}$ , and  $\varphi(B') = B_{r'}(0)$ .

**Proposition 1.11.** Every smooth manifold has a countable basis of regular coordinate balls.

*Proof.* Let  $M$  be a smooth manifold. Since  $M$  is second-countable, it admits a countable atlas  $\{(U_i, \varphi_i)\}$ . For each  $i$ , consider the collection of balls  $B_r(a) \subset \varphi_i(U_i)$  with  $a \in \mathbb{Q}^n$  and  $r \in \mathbb{Q}_{>0}$  such that there exists  $r' \in \mathbb{Q}_{>0}$  with  $r < r'$  and  $\bar{B}_r(a) \subset B_{r'}(a) \subset \varphi_i(U_i)$ . This collection is countable. For each such ball, define  $B_{i,a,r} = \varphi_i^{-1}(B_r(a))$  and  $B'_{i,a,r} = \varphi_i^{-1}(B_{r'}(a))$ . Then  $(B'_{i,a,r}, \varphi_i|_{B'_{i,a,r}})$  is a smooth chart. We have  $\varphi_i(B_{i,a,r}) = B_r(a)$ . Since  $\bar{B}_r(a)$  is compact in  $\mathbb{R}^n$ ,  $\varphi_i^{-1}(\bar{B}_r(a))$  is compact in  $M$ . Since  $M$  is Hausdorff,  $\varphi_i^{-1}(\bar{B}_r(a))$  is closed in  $M$ , so  $\varphi_i(\bar{B}_{i,a,r}) = \overline{B_r(a)}$ , and  $\varphi_i(B'_{i,a,r}) = B_{r'}(a)$ . Hence  $B_{i,a,r}$  is a regular coordinate ball. Let  $p \in M$  and let  $W \subset M$  be a neighborhood of  $p$ . Choose a chart  $(U_i, \varphi_i)$  with  $p \in U_i$ . Then  $\varphi_i(W \cap U_i)$  is open in  $\mathbb{R}^n$ . There exists a ball  $B_r(a)$  such that  $\varphi_i(p) \in B_r(a)$  and  $\bar{B}_r(a) \subset \varphi_i(W \cap U_i)$ . Pick  $r' > r$  such that  $\bar{B}_r(a) \subset B_{r'}(a) \subset \varphi_i(W \cap U_i) \cap \varphi_i(U_i)$ . Then  $p \in B_{i,a,r} \subset W$ . Hence the collection of all  $B_{i,a,r}$  forms a countable basis of  $M$ .  $\square$

**Example.** Let  $M$  be a topological 0-manifold. For every  $p \in M$ , there exists a chart  $(U, \varphi)$  and  $U \approx \{0\}$ , so  $U = \{p\}$  and  $M$  is discrete. Since  $M$  is second-countable, it has a countable basis, so there are countably many singletons in  $M$ , hence  $M$  is countable. The set of all charts  $\mathcal{A}$  is an atlas for  $M$ . Pick charts  $(\{p\}, \varphi)$  and  $(\{q\}, \psi)$ . If  $p \neq q$ , then the charts are vacuously compatible. If  $p = q$ , then  $\psi \circ \varphi^{-1} : \{0\} \mapsto \{0\}$  is the identity map, hence a diffeomorphism. Therefore  $(M, \mathcal{A})$  is a smooth manifold.  $\oslash$

**Example.** A topological manifold can admit different smooth structures. The space  $\mathbb{R}^n$  is a topological  $n$ -manifold covered by the atlas  $\{(\mathbb{R}^n, \text{id}_{\mathbb{R}^n})\}$ . The smooth structure determined by this atlas is called the *standard smooth structure* on  $\mathbb{R}^n$ . Now consider the homeomorphism  $\varphi : \mathbb{R} \rightarrow \mathbb{R}$  with  $\varphi : x \mapsto x^3$ . Cover  $\mathbb{R}$  by the atlas  $\{(\mathbb{R}, \varphi)\}$ . The transition map  $\text{id}_{\mathbb{R}} \circ \varphi^{-1}(y) = y^{1/3}$  is not smooth at 0. Hence the smooth structure determined by  $\{(\mathbb{R}, \varphi)\}$  is not the same as the standard smooth structure on  $\mathbb{R}$ .  $\oslash$

**Example.** Let  $V$  be a finite-dimensional vector space over  $\mathbb{R}$ . Any norm on  $V$  induces a metric and all norms induce the same metric topology. Let  $\{e_1, \dots, e_n\}$  be a basis of  $V$ . The map  $f_e : \mathbb{R}^n \rightarrow V$  defined by  $f_e : (x_1, \dots, x_n) \mapsto \sum x_i e_i$  is a homeomorphism, so  $(V, f_e^{-1})$  is a chart. Choosing another basis  $\{e'_i\}$  gives a chart  $(V, f_{e'}^{-1})$ . There exists a change-of-basis matrix  $(A_{ij})$  with  $e_j = \sum A_{ij} e'_i$ . We have  $f_{e'}^{-1} \circ f_e(x) = f_{e'}^{-1}((\sum_i \sum_j A_{ij} x_j) e'_i) = (\sum A_{1j} x_j, \dots, \sum A_{nj} x_j) = Ax$ , which is linear, hence smooth. The collection of those charts determines a smooth structure on  $V$ , called the *standard smooth structure* on  $V$ .  $\oslash$

**Example.** Let  $M(m \times n, \mathbb{R})$  be the set of  $m \times n$  matrices with real entries. As  $\mathbb{R}$ -vector spaces,  $M(m \times n, \mathbb{R}) \approx \mathbb{R}^{mn}$ , hence it is a smooth  $(mn)$ -manifold with the standard smooth structure.  $\oslash$

**Example.** Let  $(M, \mathcal{A})$  be a smooth  $n$ -manifold and let  $U \subset M$  be open. Consider the collection  $\mathcal{A}_U = \{(V, \varphi) \in \mathcal{A} \mid V \subset U\}$ . For every  $p \in M$ , there exists a chart  $(W, \varphi) \in \mathcal{A}$  such that  $p \in W$ . Let  $V = W \cap U$ , then  $(V, \varphi|_V) \in \mathcal{A}_U$ . Hence  $(U, \mathcal{A}_U)$  is a smooth  $n$ -manifold. Any open subset of a smooth manifold with this smooth structure is called an *open submanifold*.  $\oslash$

**Example.** The general linear group  $GL_n(\mathbb{R})$  is an open subset of  $M(n \times n, \mathbb{R})$ . Hence  $GL_n(\mathbb{R})$  is a smooth  $(n^2)$ -manifold.  $\oslash$

**Example.** Consider the set  $M_m(m \times n, \mathbb{R})$  consists of the matrix of rank  $m$ . For every  $A \in M_m(m \times n, \mathbb{R})$ , there exists a submatrix  $A' \in M(m \times m, \mathbb{R})$  of  $A$  with  $\det(A') \neq 0$ . The map  $A \mapsto \det(A')$  is continuous, there exists an open neighborhood of  $A$  in  $M(m \times n, \mathbb{R})$  where  $\det A' \neq 0$ . The matrices in this neighborhood have  $m$  linearly independent columns, so it is a neighborhood in  $M_m(m \times n, \mathbb{R})$ . Hence  $M_m(m \times n, \mathbb{R}) \subset M(m \times n, \mathbb{R})$  is open, it is a smooth  $(mn)$ -manifold.  $\oslash$

**Example.** Let  $V$  and  $W$  be finite-dimensional vector spaces over a field  $K$ . The set of linear maps from  $V$  to  $W$  form a vector space  $\mathcal{L}(V, W)$ . Hence  $\mathcal{L}(V, W)$  is a smooth  $(\dim V \cdot \dim W)$ -manifold.  $\oslash$

**Example.** Let  $\Gamma(f)$  be the graph of a smooth function  $f$ . Then  $\Gamma(f)$  is covered by the chart  $(\Gamma(f), \pi|_{\Gamma(f)})$ . It is a smooth manifold with the smooth structure  $\{(\Gamma(f), \pi|_{\Gamma(f)})\}$ .  $\oslash$

**Example.** Recall that  $S^n$  is covered by the charts  $(U_i^\pm, \varphi_i^\pm)$ , where  $\varphi_i^\pm : (x_1, \dots, x_{n+1}) \mapsto (x_1, \dots, \hat{x}_i, \dots, x_{n+1})$ . We have  $\varphi_i^\pm \circ (\varphi_j^\pm)^{-1}(x) = \varphi_i^\pm(x_1, \dots, x_{j-1}, \pm \sqrt{1 - |x|^2}, x_j, \dots, x_n) = (x_1, \dots, \hat{x}_i, \dots, \pm \sqrt{1 - |x|^2}, \dots, x_n)$  for  $i < j$ . Similarly, the cases  $i > j$  and  $i = j$  give diffeomorphisms. The case  $\varphi_i^\pm \circ (\varphi_j^\mp)^{-1}$  follows by the same argument. Hence  $S^n$  is a smooth  $n$ -manifold. The smooth structure  $\{(U_i^\pm, \varphi_i^\pm)\}$  is called the *standard smooth structure* on  $S^n$ .  $\oslash$

**Example.** Let  $U \subset \mathbb{R}^n$  be open and let  $\varphi : U \rightarrow \mathbb{R}$  be smooth. The set  $\varphi^{-1}(c)$  is called a *level set* of  $\varphi$ . Pick some level set  $M$  and suppose  $[\frac{\partial \varphi}{\partial x_1}(a), \dots, \frac{\partial \varphi}{\partial x_n}(a)] \neq 0$  for all  $a \in M$ . By the implicit function theorem, there exists a neighborhood  $U_a$  such that  $M \cap U_a$  is a graph of some smooth function  $f : V \rightarrow \mathbb{R}$ , where  $V \subset \mathbb{R}^{n-1}$  is open,

defined by  $f : (x_1, \dots, \widehat{x_i}, \dots, x_n) \mapsto x_i$ . Similar to the previous example, it is clear that  $\{(U_a \cap M, \psi_a)\}$ , where  $\psi_a : (x_1, \dots, x_n) \mapsto (x_1, \dots, \widehat{x_i}, \dots, x_n)$ , is a smooth structure on  $M$ . Hence  $M$  is a smooth  $(n-1)$ -manifold.  $\oslash$

**Example.** Recall that  $\mathbb{R}P^n$  is covered by the charts  $(U_i, \varphi_i)$ , where  $\varphi_i : \langle (x_1, \dots, x_{n+1}) \rangle \mapsto (x_1/x_i, \dots, \widehat{x_i/x_i}, \dots, x_{n+1}/x_i)$ . We have  $\varphi_i \circ \varphi_j^{-1}(x) = \varphi_i(x_1, \dots, x_{j-1}, 1, x_j, \dots, x_n) = (x_1/x_i, \dots, \widehat{x_i/x_i}, \dots, 1/x_i, \dots, x_n/x_i)$  for  $i < j$ . By the same argument, other cases give diffeomorphisms. Hence  $\mathbb{R}P^n$  is a smooth  $n$ -manifold.  $\oslash$

**Example.** Recall that the product topological manifold  $M_1 \times \dots \times M_n$  is covered by the charts  $(U_1 \times \dots \times U_n, \varphi_1 \times \dots \times \varphi_n)$ . We have  $(\varphi_1 \times \dots \times \varphi_n) \circ (\psi_1 \times \dots \times \psi_n)^{-1} = (\varphi_1 \circ \psi_1^{-1}) \times \dots \times (\varphi_n \circ \psi_n^{-1})$ . Each  $\varphi_i \circ \psi_i^{-1}$  is smooth, so the product is smooth. Hence  $M_1 \times \dots \times M_n$  is a smooth manifold.  $\oslash$

**Theorem 1.1** (chart lemma). Let  $M$  be a set. Let  $\{U_\alpha\} \subset \mathcal{P}(M)$  with maps  $\varphi_\alpha : U_\alpha \rightarrow \mathbb{R}^n$ . Then  $M$  is a smooth  $n$ -manifold with the smooth structure  $\{(U_\alpha, \varphi_\alpha)\}$  if the following properties hold.

1. For every  $\alpha$ ,  $\varphi_\alpha$  is a bijection onto an open subset of  $\mathbb{R}^n$ .
2. For every  $\alpha$  and  $\beta$ ,  $\varphi_\alpha(U_\alpha \cap U_\beta)$  and  $\varphi_\beta(U_\alpha \cap U_\beta)$  are open.
3. If  $U_\alpha \cap U_\beta \neq \emptyset$ , then the map  $\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \rightarrow \varphi_\beta(U_\alpha \cap U_\beta)$  is smooth.
4. There exist countably many  $U_\alpha$  that cover  $M$ .
5. Let  $p, q \in M$ . If  $p \neq q$ , then either there exists  $U_\alpha$  such that  $p, q \in U_\alpha$  or there exist  $U_\alpha$  and  $U_\beta$  such that  $p \in U_\alpha$ ,  $q \in U_\beta$ , and  $U_\alpha \cap U_\beta = \emptyset$ .

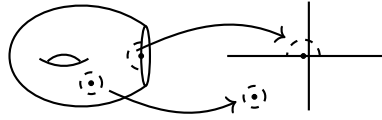
*Proof.* Define  $\{\varphi_\alpha^{-1}(V)\}$ , where  $V \subset \varphi_\alpha(U_\alpha)$  is open, as a basis of the topology. Let  $p \in \varphi_\alpha^{-1}(V) \cap \varphi_\beta^{-1}(W)$ . By (i) and (iii),  $(\varphi_\beta \circ \varphi_\alpha^{-1})^{-1}(W) \subset \varphi_\alpha(U_\alpha \cap U_\beta)$ , and by (ii),  $(\varphi_\beta \circ \varphi_\alpha^{-1})^{-1}(W)$  is open, so  $V' = V \cap (\varphi_\beta \circ \varphi_\alpha^{-1})^{-1}(W)$  is open in  $\mathbb{R}^n$ . We have  $p \in \varphi_\alpha^{-1}(V')$ . This basis induces a topology on  $M$ . Clearly (iv) implies  $M$  is second-countable and (v) implies  $M$  is Hausdorff. Hence by (iii),  $M$  is a smooth  $n$ -manifold.  $\square$

**Example.** Let  $V$  be a  $n$ -dimensional  $\mathbb{R}$ -vector space. A *Grassmannian*  $G_k(V)$  is the set of all  $k$ -dimensional linear subspaces of  $V$ . We show  $G_k(V)$  is a smooth manifold by the chart lemma. Suppose  $V = P \oplus Q$ , where  $\dim P = k$ . Given a linear map  $X : P \rightarrow Q$ , we have  $\Gamma(X) = \{v + X(v) \mid v \in P\} \subset V$ , so  $\Gamma(X) \cap Q = \{0\}$ . Let  $S \subset V$  be a  $k$ -dimensional linear subspace such that  $S \cap Q = \{0\}$ , then  $\pi_P|_S : S \rightarrow P$  is an isomorphism, so  $X = \pi_Q \circ \pi_P^{-1}$  is a well-defined linear map and  $\Gamma(X) = S$ . Let  $U_Q$  be the set of subspaces that intersect trivially with  $Q$ . Define  $\varphi : U_Q \rightarrow \mathcal{L}(P, Q)$  by  $S \mapsto X$ , so  $\varphi$  is bijective. Now  $\mathcal{L}(P, Q) \approx M((n-k) \times k, \mathbb{R}) \approx \mathbb{R}^{(n-k)k}$ , which is open. This proves (i). Let  $V = P' \oplus Q'$  be another such decomposition. Fix  $X \in \varphi(U_Q \cap U_{Q'})$ . Define  $I_X : P \rightarrow V$  by  $I_X : v \mapsto v + X(v)$ . Here  $\ker(\pi_{P'} \circ I_X) = \{0\}$ , so  $\pi_{P'} \circ I_X$  has full rank. Define  $\psi : \mathcal{L}(P, Q) \rightarrow \mathcal{L}(P, P')$  by  $\psi : X \mapsto \pi_{P'} \circ I_X$ , which is continuous. Since the matrices with full rank form an open submanifold,  $\varphi(U_Q \cap U_{Q'})$  is open. This proves (ii). Let  $X \in \varphi(U_Q \cap U_{Q'})$  and let  $S = \Gamma(X)$ . Then  $\varphi' \circ \varphi^{-1}(X) = \varphi'(\Gamma(X)) = \pi_{Q'} \circ \pi_{P'}^{-1}(X)$ . Since  $I_X$  is an isomorphism between  $P$  and  $S$ ,  $\varphi' \circ \varphi^{-1}(X) = (\pi_{Q'} \circ I_X) \circ (\pi_{P'} \circ I_X)^{-1} = (\pi_{Q'}|_P + \pi_{Q'}|_{Q'} \circ X)(\pi_{P'}|_P + \pi_{P'}|_Q \circ X)^{-1}$ . The entries of  $(\pi_{P'}|_P + \pi_{P'}|_Q \circ X)^{-1}$  are rational functions, since rational functions are infinitely differentiable,  $\varphi \circ \varphi^{-1}$  is smooth. This proves (iii). Let  $\{e_1, \dots, e_n\}$  be a basis of  $V$ , then any partition of  $n$  gives bases of  $P$  and  $Q$ . Finitely many  $U_Q$  covers  $G_k(V)$ . This proves (iv). Let  $P, P' \subset V$  be subspaces of dimension  $k$ , then there exists a subspace  $Q$  with  $\dim Q = n - k$  such that  $P \cap Q = \{0\} = P' \cap Q$ , so  $P, P' \in U_Q$ . This proves (v).  $\oslash$

**Definition 1.12.** The *closed  $n$ -dimensional upper half-space*  $\mathbb{H}^n$  is defined to be the set  $\{(x_1, \dots, x_n) \in \mathbb{R}^n \mid x_n \geq 0\}$ .

We denote the topological interior of  $\mathbb{H}^n$  to be  $\text{Int } \mathbb{H}^n$  and denote the topological boundary of  $\mathbb{H}^n$  by  $\partial \mathbb{H}^n$ . More precisely,  $\text{Int } \mathbb{H}^n = \{(x_1, \dots, x_n) \mid x_n > 0\}$  and  $\partial \mathbb{H}^n = \{(x_1, \dots, x_n) \mid x_n = 0\}$ .

**Definition 1.13.** A *topological  $n$ -manifold with boundary* is a second-countable Hausdorff space  $M$  in which every point has a neighborhood homeomorphic either to an open subset of  $\mathbb{R}^n$  or to an open subset of  $\mathbb{H}^n$ .



**Definition 1.14.** Let  $M$  be a topological  $n$ -manifold with boundary. A *chart* is a pair  $(U, \varphi)$ , where  $U \subset M$  is open and  $\varphi(U)$  is homeomorphic to either an open subset of  $\mathbb{R}^n$  or an open subset of  $\mathbb{H}^n$ . We say  $(U, \varphi)$  is an *interior chart* if  $\varphi(U)$  is open in  $\mathbb{R}^n$ . We say  $(U, \varphi)$  is a *boundary chart* if  $\varphi(U)$  is open in  $\mathbb{H}^n$  and  $\varphi(U) \cap \partial\mathbb{H}^n \neq \emptyset$ . A boundary chart with image of the form  $B_r(x) \cap \mathbb{H}^n$ , where  $x \in \partial\mathbb{H}^n$ , is called a *coordinate half-ball*.

**Definition 1.15.** Let  $M$  be a topological  $n$ -manifold with boundary. A point  $p \in M$  is an *interior point* if it is in the domain of an interior chart. The point is a *boundary point* if it is in the domain of a boundary chart  $(U, \varphi)$  such that  $\varphi(p) \in \partial\mathbb{H}^n$ . The *boundary* of  $M$ , denoted  $\partial M$ , is the set of all boundary points in  $M$ . The *interior* of  $M$ , denoted  $\text{Int}(M)$ , is the set of all interior points in  $M$ .

Notice that the interior and boundary of a topological manifold are not the same as the interior and boundary of the topological space. The open unit ball  $B^n$  has no boundary but its topological boundary is  $S^{n-1}$  if we consider  $B^n$  as a subspace of  $\mathbb{R}^n$ .

**Proposition 1.12.** Let  $M$  be a topological manifold with boundary, then  $\text{Int } M \cup \partial M = M$  and  $\text{Int } M \cap \partial M = \emptyset$ .

A self-contained proof will be shown later.

**Definition 1.16.** A *closed manifold* is a compact manifold without boundary. An *open manifold* is a non-compact manifold without boundary.

**Proposition 1.13.** Let  $M$  be a topological  $n$ -manifold with boundary. Then the following properties hold.

1. The interior  $\text{Int } M$  is open in  $M$  and it is a topological  $n$ -manifold with  $\partial(\text{Int } M) = \emptyset$ .
2. The boundary  $\partial M$  is closed in  $M$  and it is a topological  $(n-1)$ -manifold with  $\partial(\partial M) = \emptyset$ .
3. If  $n = 0$ , then  $\partial M = \emptyset$ .

*Proof.* (i) For every  $p \in \text{Int } M \subset M$ ,  $p$  is contained in the domain of some interior chart, so  $\text{Int } M$  is open and it is a topological  $n$ -manifold. For every  $p \in \text{Int } M$ ,  $p \notin \partial M$ , since  $\text{Int } M$  has dimension  $n$ ,  $p \notin \partial(\text{Int } M)$ . (ii) If  $p \in \partial M$ , then  $p$  is contained in the domain of a boundary chart. Since  $\partial M = M \setminus \text{Int } M$ ,  $\partial M$  is closed. Since  $\varphi(p) \in \partial\mathbb{H}^n$ ,  $\partial M$  is a topological  $(n-1)$  manifold. Similarly,  $\partial(\partial M) = \emptyset$ . (iii) We have  $\partial\mathbb{H}^0 = \emptyset$ , so  $\partial M = \emptyset$ .  $\square$

**Proposition 1.14.** Let  $M$  be a topological manifold with boundary. Then the following properties hold.

1. The manifold  $M$  has a countable basis of precompact coordinate balls and half-balls.
2. The manifold  $M$  is locally compact.
3. The manifold  $M$  is paracompact.
4. The manifold  $M$  is locally path-connected.
5. The manifold  $M$  has countably many components, each of which is an open subset of  $M$  and a connected topological manifold with boundary.
6. The fundamental group  $\pi_1(M)$  is countable.

*Proof.* (i) The countable cover of charts consists of interior charts and boundary charts. For interior charts, the same argument as in [Prop 1.4] applies. For boundary charts,  $\varphi_i(U_i) \subset \mathbb{H}^n$  is open. Apply the logic from [Prop 1.4] to  $B_r(x) \cap \mathbb{H}^n$ , where  $x \in \mathbb{Q}^n$ ,  $x_n = 0$ , and  $r \in \mathbb{Q}_{>0}$ . (ii) Since  $M$  is Hausdorff, by (i),  $M$  is locally compact. (iii) The proof is identical to that of [Prop 1.8]. (iv) Open balls and half balls are path-connected. The proof is identical to that of [Prop 1.5]. (v) The proof is identical to that of [Prop 1.5]. (vi) Open balls and half-balls are simply connected. The proof is identical to that of [Prop 1.9].  $\square$



**Definition 1.17.** Let  $U \subset \mathbb{H}^n$  be open. A map  $F : U \rightarrow \mathbb{R}^k$  is *smooth* if for any  $x \in U$ , there exists an open  $\tilde{U} \subset \mathbb{R}^n$  with  $x \in \tilde{U}$  and a smooth map  $\tilde{F} : \tilde{U} \rightarrow \mathbb{R}^k$  such that  $\tilde{F}|_{\tilde{U} \cap U} = F|_{\tilde{U} \cap U}$ .

We can similarly define smooth compatibility, smooth structures, and smooth manifolds with boundary.

**Proposition 1.15.** Every smooth manifold with boundary has a countable basis consisting of regular coordinate balls and half-balls.

*Proof.* The interior  $\text{Int } M$  admits a countable basis of regular coordinate balls. Let  $\{(U_i, \varphi_i)\}$  be a countable atlas for  $M$ . For boundary charts, consider the collection of half-balls  $H_r(a) = B_r(a) \cap \mathbb{H}^n$  with  $a \in \mathbb{Q}^n$ ,  $a_n = 0$ , and  $r \in \mathbb{Q}_{>0}$ , such that there exists  $r' > r$  with  $\overline{H_r(a)} \subset H_{r'}(a) \subset \varphi_i(U_i)$ . This collection is countable. Define  $B_{i,a,r} = \varphi_i^{-1}(H_r(a))$  and  $B'_{i,a,r} = \varphi_i^{-1}(H_{r'}(a))$ . Since  $H_r(a)$  is compact,  $\varphi_i^{-1}(\overline{H_r(a)})$  is closed in  $M$ . Then  $(B'_{i,a,r}, \varphi_i|_{B'_{i,a,r}})$  is a smooth boundary chart. Apply the same logic from [Prop 1.11], the union of these half-balls and the regular coordinate balls from  $\text{Int } M$  forms a countable basis for  $M$ .  $\square$

**Theorem 1.2** (chart lemma). Let  $M$  be a set. Let  $\{U_\alpha\} \subset \mathcal{P}(M)$  with maps  $\varphi_\alpha : U_\alpha \rightarrow \mathbb{R}^n$  or  $\varphi_\alpha : U_\alpha \rightarrow \mathbb{H}^n$ . Then  $M$  is a smooth  $n$ -manifold with boundary if the following properties hold.

1. For every  $\alpha$ ,  $\varphi_\alpha$  is a bijection onto an open subset of  $\mathbb{R}^n$  or  $\mathbb{H}^n$ .
2. For every  $\alpha$  and  $\beta$ ,  $\varphi_\alpha(U_\alpha \cap U_\beta)$  and  $\varphi_\beta(U_\alpha \cap U_\beta)$  are open.
3. If  $U_\alpha \cap U_\beta \neq \emptyset$ , then the map  $\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \rightarrow \varphi_\beta(U_\alpha \cap U_\beta)$  is smooth.
4. There exist countably many  $U_\alpha$  that cover  $M$ .
5. Let  $p, q \in M$ . If  $p \neq q$ , then either there exists  $U_\alpha$  such that  $p, q \in U_\alpha$  or there exist  $U_\alpha$  and  $U_\beta$  such that  $p \in U_\alpha$ ,  $q \in U_\beta$ , and  $U_\alpha \cap U_\beta = \emptyset$ .

*Proof.* The proof is similar to that of [Thm 1.1] and (i) implies that  $M$  is a manifold with boundary.  $\square$

**Proposition 1.16.** Let  $M$  be a smooth manifold with boundary and let  $U \subset M$  be open. Then the following properties hold.

1. The space  $U$  is a smooth  $n$ -manifold with boundary.
2. If  $U \subset \text{Int } M$ , then  $U$  is a smooth manifold without boundary.

*Proof.* (i) Let  $\mathcal{A}$  be a smooth atlas of  $M$ . Consider the collection  $\mathcal{A}_U = \{(V, \varphi) \in \mathcal{A} \mid V \subset U\}$ . For every  $p \in U$ , there exists a chart  $(W, \varphi) \in \mathcal{A}$  such that  $p \in W$ . Let  $V = W \cap U$ . Then  $(V, \varphi|_V) \in \mathcal{A}_U$ . Hence  $(U, \mathcal{A}_U)$  is a smooth  $n$ -manifold with boundary. (ii) Since  $\text{Int } M \subset M$  is open,  $\text{Int } M$  is a smooth submanifold without boundary. Since  $U$  is open in  $\text{Int } M$ ,  $U$  is a submanifold without boundary.  $\square$

**Proposition 1.17.** Let  $M_1, \dots, M_n$  be smooth manifolds without boundary and let  $N$  be a smooth manifold with boundary. Then  $\partial(M_1 \times \dots \times M_n \times N) = M_1 \times \dots \times M_n \times \partial N$ .

*Proof.* Let  $M = M_1 \times \dots \times M_n$ . Then  $\partial M = \emptyset$ . Let  $\dim M = k$  and  $\dim N = t$ . Since  $\partial(M \times \text{Int } N) = \emptyset$ , if  $p \in M \times \text{Int } N$ , then there exists a chart whose domain contains  $p$ . If  $p \in M \times \partial N$ , then there exists charts  $U \subset M$  and  $V \subset N$  such that  $p \in U \times V$ . Since  $U \times V$  is diffeomorphic to an open subset of  $\mathbb{R}^k \times \mathbb{H}^t \approx \mathbb{H}^{k+t}$ ,  $M \times N$  is a smooth manifold with boundary and  $\partial(M \times N) = M \times \partial N$ .  $\square$

**Proposition 1.18.** Let  $M$  be a smooth manifold with boundary and let  $p \in M$ . If there exists a boundary chart  $(U, \varphi)$  with  $p \in U$ , then any chart  $(V, \psi)$  with  $p \in V$  is a boundary chart.

*Proof.* Suppose  $\psi(p) \in \text{Int } \mathbb{H}^n$ . Then there exists a ball  $B(\psi(p)) \subset \mathbb{R}^n$  such that  $B \subset \psi(U \cap V) \subset \text{Int } \mathbb{H}^n$ . The transition map  $\tau = \varphi \circ \psi^{-1}$  is a diffeomorphism, so the Jacobian  $d\tau_x$  is invertible at every  $x \in B$ . By the inverse function theorem,  $\tau|_B$  is an open map. We have  $\tau(\psi(p)) = \varphi(p) \in \partial \mathbb{H}^n$ , so  $\tau(B)$  is an open neighborhood of  $\varphi(p)$ , a contradiction.  $\square$

## Alphabetical Index

$C^\infty$  functions, 3

$C^k$  manifold, 4

atlas, 4

boundary, 7

boundary chart, 7

boundary point, 7

centered at, 1

chart, 1, 7

closed manifold, 7

closed  $n$ -dimensional upper  
half-space, 6

coordinate ball, 1

coordinate cube, 1

coordinate domain, 1

coordinate half-ball, 7

coordinate map, 1

diffeomorphism, 3

graph, 1

Grassmannian, 6

interior, 7

interior chart, 7

interior point, 7

level set, 5

local coordinates, 1

locally finite, 3

maximal, 4

$n$ -torus, 2

open manifold, 7

open submanifold, 5

paracompact, 3

real projective space, 1

refinement, 3

regular coordinate ball, 4

smooth, 8

smooth atlas, 4

smooth chart, 4

smooth coordinate ball, 4

smooth coordinate domain, 4

smooth coordinate map, 4

smooth function, 3

smooth manifold, 4

smooth structure, 4

smoothly compatible, 3

standard smooth structure, 5

topological  $n$ -manifold, 1

topological  $n$ -manifold with  
boundary, 6

transition map, 3

## References

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